BIG DATA ANALYTICS

Course Syllabus

5 Core Units | 45 Hours | 3 Credits

A structured Big Data Analytics program covering Big Data fundamentals, Hadoop ecosystem concepts, enterprise architecture, data integration, Big Data processing technologies, real-time analytics and analytics methodology.

Course Structure

Unit	Module	Hours
Unit 1	Introduction to Big Data Analytics	9
Unit 2	Emerging Database Landscape	9
Unit 3	Application, Architecture & Data Modeling	9
Unit 4	Extracting Value from Big Data	9
Unit 5	Big Data Analytics Methodology	9

Unit 1: Introduction to Big Data Analytics

- Big Data Definition & Taxonomy
- Sources of Big Data
- 3V's of Big Data
- Need for Hadoop
- Varying Data Structures
- Characteristics of Big Data
- Applications of Big Data
- Challenges in Big Data
- Big Data Implications for Industries
- Introduction to Big Data Analytics
- Telecom, Banking, Retail, Healthcare, IT & Operations

Unit 2: Emerging Database Landscape

- Scale-Out Architecture
- RDBMS vs Non-Relational Databases
- Database Workloads
- Database Workload Characteristics
- Implications of Big Data Scale
- Introduction to Hadoop
- Hadoop Architecture
- History of Hadoop
- Hadoop Ecosystem
- Hadoop Components
- HDFS
- MapReduce

Unit 3: Application, Architecture & Data Modeling

- Big Data Warehouse & Analytics
- Big Data Warehouse System Requirements
- Hybrid Architectures
- Enterprise Data Platform Ecosystem
- Big Data and Master Data Management
- Data Integration Patterns
- ELT Pattern
- Batch Integration
- Real-Time Integration
- Data Virtualization
- Big Data Workload Design
- Analytical & Operational Workloads
- Batch & Stream Processing
- Data Partitioning Strategies
- Performance Optimization

Unit 4: Extracting Value from Big Data

- Big Data Analytics Tools
- Apache Pig
- Pig Latin
- Apache Hive
- HiveQL
- Hive Architecture
- Apache HBase
- HBase Architecture
- Apache Mahout
- Machine Learning with Big Data
- Apache Spark
- Spark Shell
- Spark Architecture
- Spark vs MapReduce
- Real-Time Analytics
- In-Memory Data Grids
- MapReduce vs Real-Time Processing
- Industry Use Cases

Unit 5: Big Data Analytics Methodology

- Introduction to Big Data Analytics Methodology
- Analyze & Evaluate Business Cases
- Develop Business Hypothesis
- Analyze Outcomes
- Build & Prepare Data Sets
- Data Collection
- Data Cleaning
- Data Integration

- Select & Build Analytical Models
- Model Evaluation
- Design for Big Data Scale
- Horizontal Scaling
- Distributed Processing
- Parallel Computing
- Production-Ready Analytics Systems
- Gathering Data
- Batch Processing
- Real-Time Streaming
- Measure & Monitor Analytics Systems
- Key Performance Indicators (KPIs)
- Healthcare Applications
- Banking Applications
- Data Quality, Data Privacy, Scalability, Infrastructure Cost & Model Bias

Technology & Concept Coverage

- Hadoop
- HDFS
- MapReduce
- Apache Pig / Pig Latin
- Apache Hive / HiveQL
- Apache HBase
- Apache Mahout
- Apache Spark / Spark Shell
- Batch Processing
- Stream Processing / Real-Time Analytics
- Big Data Architecture & Data Integration

Learning Outcomes

- Explain Big Data concepts, characteristics, sources and challenges.
- Understand Hadoop architecture and core components including HDFS and MapReduce.
- Compare database and Big Data architecture approaches for different workloads.
- Understand major Big Data technologies including Pig, Hive, HBase, Mahout and Spark.
- Apply a structured Big Data Analytics methodology from business problem definition through data preparation, model evaluation, scaling and monitoring.
- Relate Big Data Analytics concepts to industry scenarios such as banking and healthcare.

Delivery Format

45 Hours | 5 Units | 3 Credits. The program can be delivered for colleges, universities or customized corporate training. Duration, practical depth, demonstrations and project activities can be adjusted according to requirements.